

HomelyHub

AI-Powered Stay Booking Web Application

MERN Stack • JWT Auth • Groq AI • ImageKit

Internship Project

Project

An Airbnb-like app to search, book and list stays - with AI built in.

Search & Book

Find houses, flats, guest houses and hotels
for your city, dates and budget.

List Your Property

Owners upload their place with photos
and AI writes the description.

AI Trip Planner

Get a day-by-day trip plan plus
stays that fit your budget.

Problem Statement

Why we built HomelyHub

Slow Search

Finding a stay by city, budget, guests and dates takes a lot of time.

Double Booking

The same room can get booked twice for the same dates.

Hard for Owners

Small owners struggle to write an attractive property description.

Trip Planning

Planning what to do each day within budget happens on another site.

Solution

One platform for the stay AND the trip

Smart Filters

City, price, property type, room type, amenities, guests - only FREE dates shown.

Safe Booking Flow

After payment, booking is saved and dates are blocked on the house.

AI Description

Owner fills basic details -
AI writes a 3-4 sentence description.

AI Trip Planner

Destination + budget + days + interests
= day-wise plan + matching stays.

Technologies Used

MERN Stack + AI

Frontend

React (Vite)
Redux Toolkit
React Router, Axios
Ant Design, Leaflet Maps

Backend

Node.js + Express
MongoDB + Mongoose
JWT + Cookies, bcrypt
Nodemailer + Mailgen

AI & Tools

Groq SDK (LLM)
ImageKit (image upload)
Postman, Git
VS Code

Key Features

Authentication

Signup / Login / Logout
with JWT cookie

Forgot Password

Email reset link,
valid for 10 minutes

Search + Pagination

Filters and
12 properties per page

Date Availability

Already-booked dates
hidden from search

Property Details

Photos, amenities,
map, check-in time

Booking & Payment

Create order -> verify
-> confirmed

My Bookings

All trips +
booking details page

AI Features

Description writer
+ Trip Planner

My Role

Full-stack developer (change this to what YOU actually did)

Database Design

MongoDB models for
User, Property and Booking

Auth APIs

Signup, login, protect middleware,
forgot & reset password

Search API

Filters, search and pagination
using APIFeatures class

Frontend Integration

Connected React pages to APIs
using Redux and Axios

AI Integration

Description writer and
trip planner using Groq

API Testing

Tested every endpoint
in Postman

New Features / Improvements

What makes our app different

AI Description

No made-up amenities - AI uses only the details given

AI Trip Planner

JSON plan + stays where
 $\text{price per night} = \text{budget} / \text{days}$

Date Overlap Check

Search hides houses booked on the chosen dates

Secure Reset

Only the HASH of reset token is saved, expires in 10 minutes

Token Safety

Old tokens stop working after a password change

Cloud Images

Photos on ImageKit, only URL in DB + initials avatar by default

Challenges Faced

Problem -> What we did

Double Booking

Save booking AND \$push dates into
property.currentBookings

Date Overlap Logic

Clash = existing start < my check-out
AND existing end > my check-in

Password Security

bcrypt hash in pre-save hook,
password hidden with select: false

Login Lost on Refresh

/me API called on load -
cookie tells who is logged in

Cookies Across Ports

CORS with credentials +
correct sameSite / secure settings

Messy AI Output

Strict system prompt +
JSON-only response format

Results / Outcome

What We Built

A working end-to-end booking platform:
signup -> search -> book -> my trips

Impact

Owners list in minutes with AI text;
travellers plan the whole trip in one place

What I Learnt

MERN structure, REST APIs, middleware,
JWT + bcrypt security, Redux, using AI

Future Scope

Real Razorpay payment, reviews &
ratings, admin dashboard

Thank You!

Questions?

HomelyHub - search, book, list and plan your trip with AI, all in one place.